

## **PROJECT TOPIC: RECENT NATURAL DISASTERS IN THE INDIAN SUBCONTINENT**

**Topic Description:** Comprehensive Study of Recent Natural Disaster Phenomena in the Indian Subcontinent: Regional and Seasonal Profile of Two Major Disasters

The Indian subcontinent is geographically and climatologically one of the most disaster-prone regions in the world. Bounded by the tectonically active Himalayan mountain range to the north and a vast 7,500-kilometer coastline exposed to tropical cyclogenesis in the Arabian Sea and Bay of Bengal, the subcontinent experiences recurrent catastrophic hydrometeorological and geological disaster phenomena annually.

This project documents an extensive investigative and analytical study of **Recently Occurred Natural Disaster Phenomena in the Indian Subcontinent**, presenting an in-depth regional and seasonal comparative profile of two distinct major disasters: **Extremely Severe Cyclonic Storm 'Biparjoy' (Arabian Sea, June 2023)** and the **Devastating Chamoli Glacial Cloudburst & Flash Flood (Uttarakhand, February 2021)**.

The study investigates the precise meteorological and geological genesis mechanisms, seasonal patterns, geographic damage footprints, early warning systems, and socio-economic devastation inflicted by both events. By evaluating early warning forecasting and disaster response strategies, the project highlights how scientific preparedness mitigates human casualties amidst accelerating global climate change.

## 2. Importance of the topic (विषयाचे महत्त्व)

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Documenting recent natural disasters in the subcontinent provides critical insights into climate risk and disaster management:

### Key Dimensions of Disaster Studies Importance :-

- **Understanding Accelerating Climate Extremes:** Tropical cyclones in the Arabian Sea were historically rare, but rapidly warming sea surface temperatures (exceeding 30°C) are now fueling rapid intensification of severe storms, proving the direct impact of global warming on subcontinental weather systems.
- **Evaluating Early Warning Systems & Evacuation Logistics:** Cyclone Biparjoy demonstrated an extraordinary national disaster success: the India Meteorological Department (IMD) accurately forecasted the landfall trajectory 5 days in advance, enabling the timely evacuation of over 100,000 citizens with zero direct storm casualties.
- **Unpacking Fragile Himalayan Geomorphology:** The Chamoli flash flood revealed the lethal risks of building mega-hydroelectric dam tunnels in ecologically fragile, seismically active glacial zones subject to avalanche-induced lake bursts.
- **Fostering Disaster Preparedness Literacy:** Equips Standard XI students with essential scientific understanding of regional hazard zones, early warning alert color codes (Yellow, Orange, Red), and emergency reaction protocols.

Studying disasters proves that while natural hazards are inevitable, human casualties are largely preventable through scientific planning.

### 3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

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The disaster research investigation was guided by structured climatological, geographical, and risk management objectives:

#### Structured Project Objectives :-

- **Disaster Typology & Event Selection:** To select and profile two contrasting major disaster phenomena (one tropical hydrometeorological cyclone and one high-altitude geological-glacial flash flood).
- **Seasonal & Regional Genesis Analysis:** To examine the seasonal timing, atmospheric/geological origin factors, geographic path, and physical intensity of Cyclone Biparjoy and the Chamoli flood.
- **Damage & Impact Audit:** To compile quantitative data on infrastructure devastation, agricultural losses, power grid destructions, and human casualties for both disasters.
- **Early Warning & Mitigation Evaluation:** To evaluate the performance of satellite radar tracking, Doppler radar warnings, and National Disaster Response Force (NDRF) rescue deployments.
- **Climate Adaptation Action Guidelines:** To formulate practical policy recommendations for climate-resilient coastal infrastructure and Himalayan ecological conservation.

The scope encompasses IMD bulletins, NDMA post-disaster reports, and satellite imagery data.

#### 4. Project Work Methodology (प्रकल्प कार्यपद्धती)

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A standardized secondary meteorological research, cartographic review, and analytical disaster assessment methodology was deployed:

##### Methodological Protocol Executed :-

- **Phase 1 – Official Disaster Report Review:** Analyzed post-cyclone evaluation reports published by the India Meteorological Department (IMD) and National Disaster Management Authority (NDMA).
- **Phase 2 – Case Selection & Data Standardization:** Selected Cyclone Biparjoy (June 2023) and the Chamoli Disaster (February 2021); standardized empirical indicators: Genesis Mechanism, Season, Impacted Region, Wind Speed / Peak Discharge, Casualties, and Mitigation.
- **Phase 3 – Chronological Trajectory Mapping:** Tracked the 13-day lifespan of Cyclone Biparjoy across the east-central Arabian Sea to its landfall in Jakhau Port, Gujarat, and mapped the Rishi Ganga-Dhaulti Ganga flood path.
- **Phase 4 – Comparative Evaluation:** Constructed a structured comparative matrix contrasting coastal cyclonic hazards with high-altitude glacial lake outburst floods (GLOFs).
- **Phase 5 – Synthesis & Curricular Alignment:** Synthesized findings according to Chapter 7 guidelines of the Maharashtra State Board Standard XI EVS curriculum.

All recorded meteorological metrics and dates were verified against official government disaster archives.

5. Observations (निरीक्षणे)

Systematic compilation of meteorological and disaster records yielded an authoritative comparative profile of both events:

5.1 Comparative Regional & Seasonal Profile of Two Major Disasters

| Disaster Parameter             | Case Study 1: Cyclone 'Biparjoy' (June 2023)                                                                         | Case Study 2: Chamoli Flash Flood (February 2021)                                                               |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Disaster Category & Type       | Hydrometeorological: Extremely Severe Cyclonic Storm (ESCS); tropical maritime vortex.                               | Geological & Glacio-Hydrological: Giant rock-ice avalanche triggering flash flood & debris flow.                |
| Seasonal Profile & Timing      | Pre-Monsoon season (6 to 19 June 2023); longest-lived Arabian Sea cyclone (13 days).                                 | Late Winter season (7 February 2021, 10:20 AM); unexpected winter glacial detachment.                           |
| Regional Impact Zone           | East-central Arabian Sea, making landfall at Jakhau Port, Kutch (Gujarat) & south Rajasthan.                         | Rishi Ganga & Dhauli Ganga river valleys, Chamoli district, Garhwal Himalayas (Uttarakhand).                    |
| Physical Magnitude / Intensity | Sustained wind speeds of 140–150 km/h, gusting to 165 km/h; 3–4 meter coastal storm surge.                           | 27 million m³ of rock and hanging glacier broke from Ronti peak (5,600m), creating a 20m wall of mud and water. |
| Documented Impact & Damage     | 5,120 electric poles flattened, 800+ trees uprooted, 100,000+ people evacuated; zero direct storm deaths in Gujarat. | Completely washed away the 13.2 MW Rishi Ganga dam and buried 204 workers inside Tapovan dam tunnels.           |
| Key Early Warning Lesson       | Exceptional multi-agency evacuation success: accurate satellite tracking prevented catastrophic human mortality.     | Complete absence of high-altitude upstream early warning acoustic sensors in narrow Himalayan valleys.          |

## 6. Analysis of data (माहितीचे विश्लेषण)

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Analytical examination of both disasters reveals profound lessons for national disaster governance in a warming world:

### Core Analytical Inferences :-

- **Arabian Sea Thermal Warming as a Catalyst:** Cyclone Biparjoy spent an unprecedented 13 days over warm ocean waters before striking Gujarat. Historically, the Arabian Sea was cooler and hostile to cyclones; now, sea surface temperatures frequently exceed  $30.5^{\circ}\text{C}$ , providing immense thermodynamic heat energy that sustains powerful cyclones directly threatening western India and Maharashtra.
- **Triumph of Predictive Evacuation Over Brute Hazards:** The zero-fatality success in Kutch proves that while human engineering cannot halt a Category 3 cyclone, accurate multi-day predictive modeling by the IMD combined with ruthless administrative evacuation saves thousands of human lives.
- **Ecological Peril in Fragile Mountain Valleys:** The Chamoli tragedy demonstrated the severe danger of constructing heavy concrete hydroelectric dams in young, fragile Himalayan catchments. Bulldozing mountainsides and blasting tunnels beneath glaciers removes natural slope toe-support, transforming a rock-fall into an uncontrollable disaster.
- **The Need for Automated Real-Time Sensor Networks:** Over 150 workers died in the Tapovan dam tunnel because there was zero communication from upstream. Automated ultrasonic water-level sensors upstream could have provided a 15-minute warning, allowing workers to escape safely.

The analysis proves that ecological sensitivity must guide all infrastructure engineering in high-risk zones.

## 7. Results and Conclusions (निष्कर्ष)

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**Conclusion:** Recent disasters in the Indian subcontinent demonstrate the escalating reality of climate change. Mitigating future catastrophes requires embracing nature-based coastal mangrove buffers, prohibiting destructive infrastructure in fragile glacial valleys, and strengthening multi-hazard early warning networks.

### Action Checklist for Disaster Resilience :-

- ❑ Monitor official IMD and NDMA weather bulletins regularly during cyclone and monsoon seasons.
- ❑ Never venture into the sea or flooded riverbanks when Red Alerts are officially broadcast.
- ❑ Support coastal mangrove afforestation to absorb destructive cyclonic wave energy naturally.
- ❑ Advocate for stopping heavy dynamite blasting and tunneling in fragile Himalayan valleys.
- ❑ Prepare a household emergency evacuation backpack (Go-Bag) with essential supplies before monsoons.

## 8. Bibliography / References (संदर्भ सूची)

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